

The Sun

Overview

The Sun is the star at the center of the Solar System and by far its most massive member, accounting for about 99.86 % of the total mass of everything that orbits it. With a diameter of approximately 1.39 million km — about 109 times that of Earth — it is so large that roughly 1.3 million Earths could fit inside it. The Sun is classified as a G-type main-sequence star, often called a yellow dwarf, though its light appears white when viewed from space. It formed approximately 4.6 billion years ago from the gravitational collapse of a region in a giant molecular cloud, and it is roughly halfway through its main-sequence lifetime. At its core, temperatures exceed 15 million °C and pressures are so immense that nuclear fusion sustains itself continuously, releasing the energy that makes life on Earth possible.

Internal Structure

The Sun's interior is divided into distinct layers that each play a role in transporting and releasing energy. The innermost region, the core, extends to about 25 % of the solar radius and is where all nuclear energy is generated. Surrounding the core is the radiative zone, where energy moves outward extremely slowly as photons are repeatedly absorbed and re-emitted by dense plasma — a single photon may take up to 170,000 years to escape this zone. Above the radiative zone lies the convective zone, where hot plasma rises, cools at the surface, and sinks back down in convection cells called granules, each roughly the size of a small country. The visible surface, the photosphere, is a relatively thin layer about 500 km deep and has a temperature of approximately 5,500 °C. Above the photosphere are the chromosphere and the corona; the corona is a super-hot outer atmosphere reaching 1–3 million °C — a temperature inversion that scientists are still working to fully explain.

Nuclear Fusion

The Sun generates energy through nuclear fusion, primarily via the proton-proton chain reaction. In this process, four hydrogen nuclei (protons) are ultimately fused together to form one helium-4 nucleus, releasing energy in accordance with Einstein's equation $E = mc^2$. Every second, the Sun converts approximately 600 million tonnes of hydrogen into helium, releasing about 3.8×10^{26} watts of power — equivalent to the simultaneous detonation of roughly 100 billion megatonne nuclear bombs every second. Despite this enormous energy output, fusion in the core is surprisingly slow on a per-unit-volume basis; the core's power density is comparable to a compost heap. It is the sheer volume of the core that makes the total output so staggering. The Sun has consumed about half of its original hydrogen fuel and will continue fusing hydrogen for another 5 billion years before transitioning to helium fusion.

Solar Activity and Space Weather

The Sun is magnetically active, and its magnetic field drives a variety of dynamic phenomena that collectively constitute space weather. Sunspots are dark, cooler regions on the photosphere where intense magnetic fields inhibit convection; they follow an approximately 11-year solar cycle during which the number of sunspots rises and falls. Solar flares are sudden, intense bursts of radiation released when stored magnetic energy is rapidly converted to kinetic and radiant energy, and they can disrupt radio communications and GPS systems on Earth within minutes. Coronal mass ejections (CMEs) are large expulsions of magnetised plasma that can travel at up to 3,000 km/s toward Earth, triggering geomagnetic storms capable of damaging power grids and satellites. The solar wind — a continuous stream of charged particles flowing outward from the corona at 400–800 km/s — shapes the magnetospheres of all planets and carves out a vast bubble in the interstellar medium called the heliosphere, which extends well beyond the orbit of Neptune.

The Sun's Future

In approximately 5 billion years, the Sun will exhaust the hydrogen fuel in its core and begin fusing hydrogen in a shell around the inert helium core. This will cause the outer layers to expand dramatically, transforming the Sun into a red giant with a radius large enough to engulf Mercury, Venus, and possibly Earth. During the red giant phase, the Sun's luminosity will be thousands of times greater than today, effectively ending any possibility of life on the inner planets. Eventually, the Sun's outer layers will be expelled as a beautiful planetary nebula, illuminated by the hot remnant core. What remains after the nebula disperses will be a white dwarf — a dense, Earth-sized object composed primarily of carbon and oxygen, slowly cooling over billions of years. The entire sequence of stellar evolution from its current state to white dwarf is well understood and places the Sun firmly among the most common type of star in the Milky Way galaxy.